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An exact penalty difference-of-convex approach for solving inverse optimal value problems in linear programming

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ABSTRACT

This paper develops an exact penalty difference-of-convex (DC) algorithm for solving inverse optimal value problems (IOVP) in linear programming. In contrast to existing penalty-based methods that often suffer from nonconvex subproblems and lack convergence guarantees, we reformulate the original bilevel problem into a single-level program with complementarity constraints by leveraging the KKT conditions and strong duality of the forward linear program. The complementarity condition is moved into the objective through an exact ℓ_1-type penalty, which is active exactly at the points where complementarity is violated and yields a DC decomposition whose subproblems are convex programs over a fixed polyhedral feasible set. We prove that every limit point of the sequence generated by the proposed algorithm is a stationary point of the penalized problem (and, when the objective is made strongly convex by a proximal term, every accumulation point of the sequence is stationary), without relying on any constraint qualification, and we establish the exactness of the penalty: under standard conditions, global minimizers of the penalized problem coincide with those of the IOVP, and the penalty is asymptotically exact in general. Extensive numerical experiments on synthetic and high-dimensional instances, including comparisons with the bilinear-penalty reformulation approach (Lv, Chen & Wan, *Applied Mathematics Letters*, 2010) and a recent DC-based method (Li & Cui, *Mathematics of Operations Research*, 2026), together with a multiple-initialization study and a sensitivity analysis with respect to the penalty parameter, demonstrate the effectiveness, robustness, and scalability of our approach.

1. Introduction

The investigation of inverse problems spans a diverse range of disciplines, such as medical imaging [1], geophysical studies [2], optical engineering [3], acoustic modeling [4], and machine learning [5]. Within the optimization context, inverse optimization has emerged as a notable subfield, drawing increasing interest in recent years [6]. Conventional mathematical optimization takes the objective and constraint functions as inputs to produce a solution. This is often described as the "forward" optimization problem. In contrast, inverse optimization accepts a decision as input and seeks to reconstruct the objective or constraint parameters. More precisely, the aim is to identify parameter values for the forward model such that the supplied decision becomes an exact or approximate optimum. This parameter inference task is itself an optimization problem, known as the "inverse" model.

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Table 1
Comparison with previous penalty-based methods [27,28].

Feature	[27,28]	Ours
Complementarity handling	Bilinear penalty	Exact ℓ_1 DC penalty
Subproblems	Nonconvex (bilinear penalty model)	Convex (fixed polyhedral feasible set)
Convergence to stationarity	Not proved	Proved
High-dimensional experiments	No	Yes ($n = 10$, $m = 1000$)

Early systematic work on inverse optimization can be traced to Burton and Toint [7], who examined inverse versions of shortest path problems. Their findings prompted subsequent studies on inverse problems over networks [8–13]. Subsequently, the study of inverse linear programming (LP) can be traced back to the pioneering work of Zhang and Liu [14,15], who first formulated and solved inverse LP problems from a parameter estimation perspective. Their work laid the foundation for later developments on various specialized continuous inverse optimization models [16–23] and references therein.

This paper focuses on a distinct class of inverse optimization problems—the inverse optimal value problem (IOVP). In this problem, we seek a cost vector $c \in \mathbb{R}^n$ such that the optimal value of a forward linear program matches a given estimated value $\hat{v} \in \mathbb{R}$. Formally, the IOVP is formulated as

$$\min_{c} \left\{ d(v, \hat{v}) : \min_{x}\{c^T x \mid Ax \geq b\} = v,\ c \in C \right\}, \tag{1}$$

where $d(v, \hat{v})$ denotes a distance measure between v and $\hat{v}$, $c \in \mathbb{R}^n$ is the cost vector, $x \in \mathbb{R}^n$ is the decision variable, $A \in \mathbb{R}^{m\times n}$ is the constraint matrix, $b \in \mathbb{R}^m$ is the right-hand side vector, and $C \subset \mathbb{R}^n$ denotes the admissible polyhedral set of cost vectors. Problem (1) was introduced by Ahmed and Guan [24] and has since been studied in various forms [25–28].

Existing methods for IOVP often rely on penalty approaches or duality-based reformulations, but face challenges in penalty parameter tuning and convergence guarantees. The closest predecessors [27,28] employ penalty functions to handle complementarity constraints derived from the KKT conditions of the forward LP. However, they suffer from two major limitations:

1. The penalty terms in [27,28] involve products $y_i s_i$ of the dual and slack variables, which destroy the convexity of the resulting subproblems and make them difficult to solve globally.
2. No convergence guarantee to stationary points is provided and the algorithms are purely heuristic.

In contrast, our work moves the complementarity constraints into the objective through an exact ℓ_1-type penalty and exploits its natural DC decomposition (see (7) in Section 2). The feasible region of the resulting DC program is a polyhedron, so every subproblem generated by our algorithm is a convex program solvable by off-the-shelf solvers such as IPOPT [29]. Moreover, we prove that every limit point of the sequence generated by the proposed algorithm is a stationary point of the penalized problem (and, when the objective is made strongly convex by a proximal term, every accumulation point is stationary), and we establish the exactness of the penalty: under standard conditions, global minimizers of the penalized problem coincide with those of the IOVP. Table 1 summarizes the key theoretical and numerical differences.

Very recently, Li and Cui [30] proposed a variational framework and a DC-type algorithm for a class of asymptotically approachable nonconvex problems, which can also be applied to the IOVP. Their work relies on epi-convergence and provides convergence guarantees under certain conditions. In this paper, we also include numerical comparisons with their method to further highlight the advantages of our exact penalty DC decomposition.

To overcome the aforementioned difficulties, we propose a penalty-type difference-of-convex algorithm (DCA). Our main contributions are threefold:

- **DC reformulation with exact penalty.** We reformulate the bilevel IOVP as a single-level program with complementarity constraints using the KKT conditions and strong duality of the lower level of problem (1). The complementarity condition is then moved into the objective through an exact ℓ_1-type penalty, leading to the following penalty-type model:

$$\begin{aligned} \min_{c,x,y,s}\quad & d(y^T b, \hat{v}) + \rho\, e^T \left(y - \Pi_{\mathbb{R}^m_+}(y - s) \right) \\ \text{s.t.}\quad & c - A^T y = 0,\ c \in C, \\ & Ax - b = s, \\ & y \in \mathbb{R}^m_+,\ s \in \mathbb{R}^m_+, \end{aligned} \tag{2}$$

  where $e := (1, 1, \ldots, 1)^T \in \mathbb{R}^m$, $\rho > 0$ is the penalty parameter, and $\Pi_{\mathbb{R}^m_+}$ denotes the projection onto $\mathbb{R}^m_+$ (i.e., $(\Pi_{\mathbb{R}^m_+}(w))_i = \max\{w_i, 0\}$). The penalty term $\rho e^T(y - \Pi_{\mathbb{R}^m_+}(y - s)) = \rho \sum_{i=1}^m \min\{y_i, s_i\}$ vanishes exactly on the complementarity set $\{(y, s) \in \mathbb{R}^m_+ \times \mathbb{R}^m_+ : y^T s = 0\}$ and is positive at every point where complementarity is violated, it is therefore genuinely needed in the formulation, and under the exactness conditions established in Theorem 2.1, a finite value of ρ suffices. Throughout this paper, we make the following standing assumptions:

(AS1) The set $C \subset \mathbb{R}^n$ is nonempty, convex, compact and polyhedral;
(AS2) The distance function $d(v, \hat{v})$ is convex, coercive and continuous in v for a fixed $\hat{v}$;
(AS3) The strong duality of the lower level of the IOVP holds.

- **Convergence guarantee.** We prove that every limit point of the sequence generated by the proposed DCA is a stationary point of the penalized problem (Theorem 3.1), with every accumulation point stationary whenever the objective is made strongly convex, and we establish the exactness of the penalty (Theorem 2.1), so that, under standard conditions, solutions of the penalized model solve the inverse optimal value problem (1).
- **Numerical validation.** We demonstrate the effectiveness of our method through extensive synthetic and high-dimensional experiments, including comparisons with the bilinear-penalty reformulation approaches [27,28] and the recent method of Li and Cui [30].

The paper is structured as follows. Section 2 presents the DC algorithm for the penalty-type variant (4). Section 3 establishes the convergence results. Section 4 reports numerical experiments on synthetic and high-dimensional problems. Finally, concluding remarks are given in Section 5.

2. Algorithm

This section aims to propose a difference-of-convex algorithm (DCA) to solve a single-level reformulation of the inverse optimal value problem (1).

By the KKT conditions of the forward problem (1) and Assumption (AS3), the IOVP is equivalent to the following single-level program with complementarity constraints

$$\begin{array}{ll} \min_{c,x,y,s} & d(y^T b, \hat{v}) \\ s.t. & c - A^T y = 0,\ c \in C, \\ & Ax - b = s, \\ & y \in \mathbb{R}^m_+,\ s \in \mathbb{R}^m_+,\ y^T s = 0. \end{array} \tag{3}$$

Indeed, a point (c, x, y, s) is feasible for (3) if and only if x is feasible for the forward problem, y is feasible for its dual, and complementary slackness holds; by strong duality this is equivalent to x being an optimal solution of the forward problem with cost vector c, in which case $y^T b$ is the corresponding optimal value. The objective of (3) therefore coincides with $d(v, \hat{v})$ in (1). Since the constraint $y^T s = 0$ is nonconvex, we penalize its violation in the objective. The following lemma collects the elementary properties of the complementarity system used in this paper.

Lemma 2.1. *For every $z := (c, x, y, s)$ satisfying $c - A^T y = 0$, $Ax - b = s$, $y \in \mathbb{R}^m_+$ and $s \in \mathbb{R}^m_+$, we have*

$$c^T x - y^T b = y^T s.$$

Consequently, with $p(z) := e^T\left(y - \Pi_{\mathbb{R}^m_+}(y - s)\right) = \sum_{i=1}^m \min\{y_i, s_i\}$, one has $p(z) \geq 0$ and

$$p(z) = 0 \iff y^T s = 0.$$

Proof. The identity $c^T x = (A^T y)^T x = y^T(Ax) = y^T(b + s) = y^T b + y^T s$ gives $c^T x - y^T b = y^T s$. Since $y_i, s_i \geq 0$, we have $\min\{y_i, s_i\} \geq 0$, and $\min\{y_i, s_i\} = 0$ if and only if $y_i s_i = 0$. Moreover, $\min\{y_i, s_i\} = y_i - \max\{y_i - s_i, 0\} = y_i - \Pi_{\mathbb{R}_+}(y_i - s_i)$, which yields the formula for $p(z)$. □

Remark 2.1. Lemma 2.1 shows that, on the linear manifold defined by $c = A^T y$, $Ax - b = s$ and $y, s \geq 0$, the quantity $c^T x - y^T b$ equals the nonnegative number $y^T s$. Hence any formulation that imposes the inequality $c^T x - y^T b \leq 0$ on this manifold automatically enforces complementarity, and a penalty term that vanishes identically on the feasible set of such a formulation has no influence on the model. This is the reason we do not keep $c^T x - y^T b \leq 0$ as a constraint. Instead, the function p in Lemma 2.1 measures precisely the amount by which complementarity is violated at points of the polyhedron

$$D := \{z = (c, x, y, s) : c - A^T y = 0,\ Ax - b = s,\ c \in C,\ y \in \mathbb{R}^m_+,\ s \in \mathbb{R}^m_+\},$$

and it is positive exactly on $D \setminus \{z : y^T s = 0\}$. The penalized problem formulated below therefore allows complementarity to be violated at intermediate iterates, and the penalty is active exactly there; this gives the penalty a genuine, nontrivial role in the formulation.

Remark 2.2. A natural alternative is to keep $c^T x - y^T b \leq 0$ as a constraint and to exploit the DC representation $c^T x = \frac{1}{2}\|c + x\|^2 - \frac{1}{2}\|c\|^2 - \frac{1}{2}\|x\|^2$ of the bilinear term. We emphasize that such a constraint-based DC decomposition is degenerate for the IOVP. Indeed, for the decomposition $g_2(z) := \frac{1}{2}\|c + x\|^2$ and $h_2(z) := \frac{1}{2}(\|c\|^2 + \|x\|^2) + y^T b$, a direct computation using Lemma 2.1 gives, for every $z \in D$ and every linearization point $w = (w_c, w_x, w_y, w_s)$,

$$g_2(z) - h_2(z; w) = y^T s + \tfrac{1}{2}\|c - w_c\|^2 + \tfrac{1}{2}\|x - w_x\|^2 \geq 0,$$

where $h_2(\cdot; w)$ denotes the first-order approximation of h_2 at w. The convexified constraint $g_2(z) - h_2(z; w) \leq 0$ can therefore never be strict on D; it forces $c = w_c$, $x = w_x$ and $y^T s = 0$. The resulting subproblems are then either infeasible (when no $y \geq 0$ with $A^T y = w_c$ is complementary to $s = Ax_0 - b$) or freeze c and x at the current iterate, so that the algorithm cannot change the recovered cost vector. This is why we penalize complementarity in the objective rather than convexifying the bilinear constraint.

We now introduce the penalized problem. Given the complementarity measure p of Lemma 2.1 and a penalty parameter $\rho > 0$, consider

$$\begin{array}{ll} \min_z & d(y^T b, \hat{v}) + \rho\, p(z) \\ \text{s.t.} & z \in D, \end{array} \tag{4}$$

where D is the polyhedron defined in Remark 2.1, or equivalently

$$D := \{z : \hat{A}z = \hat{b},\ z \in \Omega\}, \quad \hat{A} = \begin{bmatrix} I & 0 & -A^T & 0 \\ 0 & A & 0 & -I \end{bmatrix}, \quad \hat{b} = \begin{bmatrix} 0 \\ b \end{bmatrix}, \quad \Omega = C \times \mathbb{R}^n \times \mathbb{R}^m_+ \times \mathbb{R}^m_+. \tag{5}$$

Problem (4) is a relaxation of (3) in which complementarity is enforced by the penalty; the feasible region D is polyhedral and does not depend on ρ.

The following theorem establishes the exactness of the penalty. Parts (i) and (ii) give global exactness in the asymptotic and finite regimes, respectively, and part (iii) records the classical local exactness property of the ℓ_1 penalty for programs with complementarity constraints.

Theorem 2.1. *Suppose that Assumptions (AS1)–(AS3) hold, the feasible set of (3) is nonempty, and the level sets $\{z \in D : d(y^T b, \hat{v}) \le \alpha\}$ are bounded for every $\alpha \in \mathbb{R}$.*

- *(i) (Asymptotic exactness) Let $\{\rho_k\}$ be a nondecreasing sequence with $\rho_k \to +\infty$, and let $z^k := (c^k, x^k, y^k, s^k)$ be a global optimal solution of problem (4) with $\rho = \rho_k$. Then every accumulation point $\bar{z} := (\bar{c}, \bar{x}, \bar{y}, \bar{s})$ of $\{z^k\}$ is a global optimal solution of problem (3); consequently $(\bar{c}, \bar{x})$ is a global optimal solution of problem (1).*
- *(ii) (Finite exactness) Assume in addition that there exists $\kappa > 0$ with $\operatorname{dist}(z, S) \le \kappa\, p(z)$ for all $z \in D$, where $S := \{z \in D : y^T s = 0\}$ denotes the feasible set of (3), and that $z \mapsto d(y^T b, \hat{v})$ is Lipschitz continuous on D with constant L. Then for every $\rho > L\kappa$, every global optimal solution of problem (4) is a global optimal solution of the IOVP (1).*
- *(iii) (Local exactness) If z^* is a local optimal solution of problem (3) at which the MPEC-LICQ condition holds, then there exists $\bar{\rho} > 0$ such that z^* is a local optimal solution of problem (4) for every $\rho \ge \bar{\rho}$.*

Proof. Let v^* be the optimal value of problem (3) and let z^* be its global optimal solution. For every k, the point z^* is feasible for problem (4) with $\rho = \rho_k$ (since $p(z^*) = 0$), so

$$d((y^k)^T b, \hat{v}) + \rho_k\, p(z^k) \le d((y^*)^T b, \hat{v}) = v^*.$$

Since $d \ge 0$ and $p \ge 0$, this gives $d((y^k)^T b, \hat{v}) \le v^*$ and $p(z^k) \le v^*/\rho_k$. By the boundedness of the level sets of d on D, the sequence $\{z^k\}$ is bounded; for any convergent subsequence $z^k \to \bar{z}$ we obtain $p(\bar{z}) = 0$ and $d(\bar{y}^T b, \hat{v}) \le v^*$. As $\bar{z} \in D$ and $p(\bar{z}) = 0$, $\bar{z}$ is feasible for (3), so $d(\bar{y}^T b, \hat{v}) \ge v^*$. Hence equality holds and $\bar{z}$ is a global optimal solution of (3). By Lemma 2.1 and Assumption (AS3), $\bar{x}$ is an optimal solution of the forward problem with cost vector $\bar{c}$ and $\bar{y}^T b$ is the corresponding optimal value, so $(\bar{c}, \bar{x})$ solves (1). This proves (i).

For (ii), let $\rho > L\kappa$ and let z be a global optimal solution of (4). Choose $z' \in S$ with $\|z - z'\| = \operatorname{dist}(z, S)$; since $d((y')^T b, \hat{v}) \ge v^*$ for every $z' \in S$, the Lipschitz continuity of d gives $d(y^T b, \hat{v}) \ge v^* - L\|z - z'\|$, and the error bound gives $\|z - z'\| \le \kappa\, p(z)$; hence

$$d(y^T b, \hat{v}) + \rho\, p(z) \ge v^* + (\rho - L\kappa)\, p(z) \ge v^*.$$

Since $z^* \in S$ attains the value v^* in problem (4), the last inequality must be an equality at the minimizer z, which forces $p(z) = 0$. Thus $z \in S$ and $d(y^T b, \hat{v}) = v^*$, so z solves the IOVP.

Part (iii) is the classical exactness property of the ℓ_1 penalty for mathematical programs with complementarity constraints under MPEC-LICQ; see [31,32] for more details. □

Remark 2.3. Theorem 2.1 concerns the exactness of the penalized model (4): it specifies when and in which sense global solutions of the penalized problem are meaningful for the original IOVP. The DCA proposed below does not solve (4) to global optimality in general; it generates a sequence of points that are stationary for the penalized problem (Section 3). The numerical experiments in Section 4, including multiple random initializations, indicate that the stationary points found by the algorithm are consistent and close to global solutions on the tested instances.

To leverage the theory of DC optimization [33,34], we rewrite problem (4) as a DC program

$$\begin{array}{ll} \min_z & g_1(z) - h_1(z) \\ s.t. & z \in D, \end{array} \tag{6}$$

where $z := (c, x, y, s)$ and the functions g_1 and h_1 are defined as

$$g_1(z) := d(y^T b, \hat{v}) + \rho\, e^T y, \quad h_1(z) := \rho\, e^T \Pi_{\mathbb{R}^m_+}(y - s). \tag{7}$$

Both g_1 and h_1 are convex functions, so problem (6) is a DC program over the polyhedron D defined in (5).

Analogous to the DCA framework presented in [33], we consider the first-order approximation of the function $h_1(z)$ at the point $\tilde{z} := (\tilde{c}, \tilde{x}, \tilde{y}, \tilde{s})$,

$$h_1(z; \tilde{z}) := h_1(\tilde{z}) + \tilde{V}^T (z - \tilde{z}), \tag{8}$$

where $\tilde{V} := (0, 0, \tilde{V}_y, \tilde{V}_s) \in \partial h_1(\tilde{z})$. In particular, one may take $\tilde{V}_y = \rho u,\ \tilde{V}_s = -\rho u$ with $u \in \partial \Pi_{\mathbb{R}^m_+}(\tilde{y} - \tilde{s}) \subset [0, 1]^m$. By the convexity of h_1, it holds

$$h_1(z; \tilde{z}) \le h_1(z), \quad h_1(\tilde{z}; \tilde{z}) = h_1(\tilde{z}). \tag{9}$$

The subproblem corresponding to problem (6) at the point $\tilde{z}$ takes the following form

$$\begin{array}{ll} \min_z & g_1(z) - h_1(z; \tilde{z}) \\ s.t. & z \in D, \end{array} \tag{10}$$

where $g_1(z)$, $h_1(z)$, $h_1(z; \tilde{z})$ are defined as in (7), (8) and D is defined as in (5). Problem (10) is a convex optimization problem, denoted by CP($\tilde{z}, \tilde{V}$); its feasible region is the fixed polyhedron D, which does not depend on $\tilde{z}$ or $\tilde{V}$. Consequently, every subproblem is feasible whenever D is nonempty, and the penalty parameter ρ does not affect the feasibility of the subproblems. To end this section, we present a DCA for solving problem (6).

Algorithm 1 DCA for problem (6).

1: **Input:** $\rho > 0$, $z^0 \in D$, tolerance $\varepsilon > 0$, set $k = 0$, where D is the polyhedron defined in (5).
2: **repeat**
3: Select $V^k \in \partial h_1(z^k)$;
4: Solve the convex subproblem CP(z^k, V^k) to obtain z^{k+1};
5: Set $k = k + 1$.
6: **until** $\|z^{k+1} - z^k\|_2 \le \varepsilon$
7: **Output:** z^{k+1}

3. Convergence analysis

This section analyzes the relationship between the optimal solution of problem (10) and the stationary point of problem (6), and then presents the corresponding convergence results for Algorithm 1.

Definition 3.1. A point $z^* \in D$ is called a stationary point of problem (6) (equivalently, of the penalized problem (4)) if

$$0 \in \partial g_1(z^*) - \partial h_1(z^*) + \mathcal{N}_D(z^*),$$

where g_1, h_1 are defined as in (7), D is defined as in (5), and $\mathcal{N}_D(z^*)$ denotes the normal cone of the convex set D at z^*.

Since D is polyhedral, its normal cone admits an explicit representation. We record it here because it makes the stationarity system of Definition 3.1 fully explicit.

Lemma 3.1. *For every $z = (c, x, y, s) \in D$, the normal cone of the polyhedron D at z is given by*

$$\mathcal{N}_D(z) = \hat{A}^T \mathbb{R}^{m+n} + \mathcal{N}_C(c) \times \{0\} \times \left\{\lambda \in \mathbb{R}^m_- : \lambda_i = 0 \text{ whenever } y_i > 0\right\} \times \left\{\mu \in \mathbb{R}^m_- : \mu_i = 0 \text{ whenever } s_i > 0\right\},$$

where $\hat{A}$ and Ω are defined as in (5).

Proof. The set D is the intersection of the affine subspace $\{z : \hat{A}z = \hat{b}\}$ and the box-like set Ω. For polyhedral convex sets, the normal cone of an intersection is the sum of the normal cones of the two sets (see [35]), i.e., $\mathcal{N}_D(z) = \hat{A}^T \mathbb{R}^{m+n} + \mathcal{N}_\Omega(z)$. The normal cone of $\Omega = C \times \mathbb{R}^n \times \mathbb{R}^m_+ \times \mathbb{R}^m_+$ is the product of the normal cones of its factors, which yields the stated formula. □

Lemma 3.2. *If $\tilde{z}$ is an optimal solution of problem CP($\tilde{z}, \tilde{V}$) for some $\tilde{V} = (0, 0, \tilde{V}_y, \tilde{V}_s) \in \partial h_1(\tilde{z})$, then $\tilde{z}$ is a stationary point of problem (6) in the sense of Definition 3.1.*

Proof. The subproblem CP($\tilde{z}, \tilde{V}$) is a convex program over the polyhedron D. By the classical optimality condition for convex optimization, $\tilde{z}$ is an optimal solution if and only if

$$0 \in \partial g_1(\tilde{z}) - \tilde{V} + \mathcal{N}_D(\tilde{z}).$$

No constraint qualification is needed because the feasible set is exactly the polyhedron D. Since $\tilde{V} \in \partial h_1(\tilde{z})$, the inclusion $\partial g_1(\tilde{z}) - \tilde{V} \subset \partial g_1(\tilde{z}) - \partial h_1(\tilde{z})$ gives the conclusion. □

Lemma 3.3. *Let $\{z^k\}$ be generated by Algorithm 1, and let $f(z) := g_1(z) - h_1(z)$ denote the objective function of problem (6). Then the sequence $\{f(z^k)\}$ is monotonically nonincreasing and convergent.*

Proof. Since $z^{k+1} \in D$ solves CP(z^k, V^k), and $z^k \in D$ is feasible for that subproblem (its feasible region is the fixed set D), we have

$$f(z^{k+1}) = g_1(z^{k+1}) - h_1(z^{k+1}) \le g_1(z^{k+1}) - h_1(z^{k+1}; z^k) \le g_1(z^k) - h_1(z^k) = f(z^k), \tag{11}$$

where the first inequality follows from (9), the second inequality follows from the optimality of z^{k+1} in CP(z^k, V^k), and the equalities follow from $h_1(z^k; z^k) = h_1(z^k)$. Since $f(z) = d(y^T b, \hat{v}) + \rho p(z) \ge 0$ on D, the sequence $\{f(z^k)\}$ is bounded below; hence it is convergent. □

We emphasize that no constraint qualification of any kind is needed in what follows: the feasible region of every subproblem is the fixed polyhedron D, so the optimality conditions for the convex subproblems hold automatically.

Lemma 3.4. *Suppose that Assumptions (AS1)–(AS3) hold and that the level sets $\{z \in D : f(z) \le \alpha\}$ of the penalized objective are bounded for every $\alpha \in \mathbb{R}$. Then the sequence $\{z^k\}$ generated by Algorithm 1 is bounded.*

Proof. By Lemma 3.3, $f(z^k) \le f(z^0)$ for every k, so $\{z^k\} \subset \{z \in D : f(z) \le f(z^0)\}$, which is bounded by assumption. □

With the boundedness of the sequence established in Lemma 3.4, we now state the main convergence result of the algorithm.

Theorem 3.1. *Let $\{z^k\}$ be generated by Algorithm 1 under the assumptions of Lemma 3.4, and suppose that every subproblem $\mathrm{CP}(z^k, V^k)$ admits an optimal solution. Then: (i) if $\{z^k\}$ converges to $\bar{z}$, then $\bar{z}$ is a stationary point of problem (6) in the sense of Definition 3.1; (ii) if, in addition, g_1 is strongly convex on D with modulus $\mu > 0$, then $\|z^{k+1} - z^k\| \to 0$ and every accumulation point of $\{z^k\}$ is a stationary point of problem (6).*

Proof. Since z^{k+1} solves the convex program $\mathrm{CP}(z^k, V^k)$ over the polyhedron D, the optimality condition

$$0 \in \partial g_1(z^{k+1}) - V^k + \mathcal{N}_D(z^{k+1}) \tag{12}$$

holds for every k; hence there exist $u^k \in \partial g_1(z^{k+1})$ and $v^k \in \mathcal{N}_D(z^{k+1})$ with $u^k - V^k + v^k = 0$. Since $\{z^k\}$ converges to $\bar{z}$, so does $\{z^{k+1}\}$. The sequence $\{V^k\}$ is bounded because $V^k = \rho(0, 0, u_y^k, -u_y^k)$ with $u_y^k \in \partial \Pi_{\mathbb{R}^m_+}(y^k - s^k) \subset [0,1]^m$; passing to a subsequence if necessary, $V^k \to \bar{V}$, and the outer semicontinuity of the subdifferential of the finite convex function h_1 yields $\bar{V} \in \partial h_1(\bar{z})$. The sequences $\{u^k\}$ and $\{v^k\}$ are bounded (for $\{v^k\}$, this follows from $v^k = V^k - u^k$), so passing to further subsequences gives $u^k \to \bar{u}$ and $v^k \to \bar{v}$ with $\bar{u} \in \partial g_1(\bar{z})$ by the outer semicontinuity of the subdifferential of the continuous convex function g_1, and $\bar{v} \in \mathcal{N}_D(\bar{z})$ by the outer semicontinuity of the normal cone mapping of the closed convex set D. Taking limits in $u^k - V^k + v^k = 0$ yields

$$0 \in \partial g_1(\bar{z}) - \bar{V} + \mathcal{N}_D(\bar{z}) \subseteq \partial g_1(\bar{z}) - \partial h_1(\bar{z}) + \mathcal{N}_D(\bar{z}),$$

so $\bar{z}$ is a stationary point of problem (6). This proves (i).

To prove (ii), suppose that g_1 is strongly convex on D with modulus $\mu > 0$. Combining the strong convexity of g_1 with the optimality of z^{k+1} in $\mathrm{CP}(z^k, V^k)$ yields the standard DCA descent estimate

$$f(z^k) - f(z^{k+1}) \ge \tfrac{\mu}{2}\|z^{k+1} - z^k\|^2.$$

Summing over k and using $f \ge 0$ on D gives $\sum_{k\ge 0} \|z^{k+1} - z^k\|^2 \le \frac{2}{\mu} f(z^0) < +\infty$, hence $\|z^{k+1} - z^k\| \to 0$. Let $\bar{z}$ be an accumulation point of $\{z^k\}$ and let $\{z^{k_j}\}$ be a subsequence converging to $\bar{z}$ (it exists by Lemma 3.4); then $z^{k_j+1} \to \bar{z}$ as well. The sequences $\{V^{k_j}\}$, $\{u^{k_j}\}$ and $\{v^{k_j}\}$ are bounded — $\{V^{k_j}\}$ because $V^k = \rho(0, 0, u_y^k, -u_y^k)$ with $u_y^k \in \partial \Pi_{\mathbb{R}^m_+}(y^k - s^k) \subset [0,1]^m$, and $\{u^{k_j}\}$ and $\{v^{k_j}\}$ because ∂g_1, ∂h_1 and $\mathcal{N}_D$ are locally bounded on the bounded set $\{z \in D : f(z) \le f(z^0)\}$ — so, passing to further subsequences, $u^{k_j} \to \bar{u} \in \partial g_1(\bar{z})$, $V^{k_j} \to \bar{V} \in \partial h_1(\bar{z})$ and $v^{k_j} \to \bar{v} \in \mathcal{N}_D(\bar{z})$, where the outer semicontinuity is applied at $z^{k_j+1} \to \bar{z}$ for ∂g_1 and $\mathcal{N}_D$ and at $z^{k_j} \to \bar{z}$ for ∂h_1. Taking limits in $u^{k_j} - V^{k_j} + v^{k_j} = 0$ yields $0 \in \partial g_1(\bar{z}) - \partial h_1(\bar{z}) + \mathcal{N}_D(\bar{z})$, i.e., $\bar{z}$ is a stationary point of problem (6). This proves (ii). □

Remark 3.1. Theorem 3.1 is the classical DCA convergence result (see [33,34]) specialized to the polyhedral feasible region D; in particular, no constraint qualification appears in it. In the implementation, the algorithm is stopped when $\|z^{k+1} - z^k\|_2 \le \varepsilon$; at the final iterate the stationarity system (12) is satisfied up to a residual that vanishes as $\varepsilon \to 0$: indeed, along a terminating sequence with $\|z^{k+1} - z^k\| \le \varepsilon$, the inclusion $0 \in \partial g_1(z^{k+1}) - V^k + \mathcal{N}_D(z^{k+1})$ and the outer semicontinuity of ∂g_1, ∂h_1 and $\mathcal{N}_D$ imply that the distance from 0 to $\partial g_1(z^{k+1}) - \partial h_1(z^{k+1}) + \mathcal{N}_D(z^{k+1})$ tends to zero as $\varepsilon \to 0$. Moreover, whenever g_1 is strongly convex on D — which can be ensured by adding a small proximal term to the objective, for instance $(\alpha/2)\|z - z^0\|^2$ — one has $f(z^k) - f(z^{k+1}) \ge \frac{\mu}{2}\|z^{k+1} - z^k\|^2$ with $\mu > 0$, so $\|z^{k+1} - z^k\| \to 0$ and Theorem 3.1(ii) applies to every accumulation point of $\{z^k\}$.

Remark 3.2. Together with Theorem 2.1, Theorem 3.1 provides the complete convergence picture of the proposed approach. For any fixed value of ρ, Algorithm 1 generates a decreasing sequence of penalized objective values and converges to a stationary point of the penalized problem (4), without any constraint qualification and with every subproblem convex and feasible. When ρ is chosen sufficiently large, the exactness properties of Theorem 2.1 guarantee that global minimizers of the penalized problem solve the IOVP; on the tested instances, the stationary points produced by the algorithm are close to such global minimizers (see Section 4).

4. Numerical experiments

4.1. Experimental setup

This section presents numerical experiments for Algorithm 1 on synthetic and high-dimensional instances, together with comparisons against the bilinear-penalty reformulation approaches [27,28] and a recent DC-based method [30]. All codes were implemented in Python 3.14 with NumPy 2.5 and SciPy 1.18 [36]. The forward linear programs of the lower level and the linear subproblem in the variable x were solved with the HiGHS solver [37]; the convex quadratic subproblems in the variables (y, t) (the epigraph variable t is used only for the nonsmooth distance d_1) were solved with the interior-point trust-region method of SciPy, using the analytic gradients and Hessian–vector products derived in Section 2. Because the polyhedron D factorizes as

$\{c = A^T y,\ y \geq 0,\ c \in C\} \times \{s = Ax - b,\ s \geq 0\}$, subproblem (10) separates into a convex quadratic block in (y, t) and a linear block in x, which are solved separately by the two solvers above. The experiments were conducted on a computer with an Intel(R) Core(TM) i7-7700 CPU @ 3.60GHz processor and 16.00 GB of RAM.

In all experiments the penalty parameter is updated as $\rho_k := \rho_{k-1} + 10$ with initial value $\rho_0 = 10^3$, and the proximal weight of Remark 3.1 is $\alpha = 2 \times 10^{-2}$. Algorithm 1 is terminated when the step-size criterion $\|z^{k+1} - z^k\|_2 \leq 10^{-4}$ holds (for the synthetic example we use the tighter value 10^{-6}, because the two-dimensional problem converges very quickly), with at most 300 iterations (400 for the synthetic example) and a time budget of 900 seconds per run. Every iterate produced by the solver lies in the polyhedron D up to the feasibility tolerance: the equalities $c = A^T y$ and $Ax - b = s$ hold by construction, and the box and nonnegativity constraints were satisfied to 10^{-12} by the trust-region solver in all reported runs. Complementarity is not imposed as a constraint: intermediate iterates may violate it, the violation being driven to zero by the exact penalty. At a tie $y_i = s_i$ the subgradient element $u_i \in \partial \Pi_{\mathbb{R}_+}(y_i - s_i)$ was fixed to the midpoint $1/2$ (any element of $[0, 1]$ is admissible, since $\partial \Pi_{\mathbb{R}_+}(0) = [0, 1]$). The proximal term $(\alpha/2)\|c - c^0\|^2$ of Remark 3.1 selects, among the points of the value-matching set, the one closest to the initial estimate c^0; it vanishes from the model as $\alpha \to 0$ and does not change the value-matching set. In the tables below, "Res" denotes $\|c^* - \hat{c}\|_2$, "d" the value-matching residual $|Q(c^*) - \hat{v}|$ at the final iterate, "p" the complementarity measure $p(z^*)$, "Time(s)" the computation time in seconds, and "Iter" the number of DCA iterations.

4.2. Comparison with the bilinear-penalty reformulation approaches [27,28]

In this subsection, we present numerical comparisons with the classical *bilinear-penalty* reformulation of the KKT conditions used in [27,28], in which the complementarity condition is penalized by the product $y^T s$, giving

$$\min_{z \in D} \left\{ d_i(y^T b, \hat{v}) + \rho\, y^T s + \tfrac{\alpha}{2}\|c - c^0\|^2 \right\},$$

which is a nonconvex program (the bilinear term destroys convexity). We solve it with the block-coordinate descent method (BCD-BP) that alternates between the convex block in y (with s fixed) and the linear program in x (with y fixed), using the same penalty schedule, proximal weight, and stopping rules as for Algorithm 1; this is a direct realization of the penalty-type approaches of Lv et al. [27,28]. In the synthetic example we also report the generic DC algorithm applied to the bilinear-penalty model (DCA-BP), i.e., the DC decomposition $y^T s = \frac{1}{4}\|y + s\|^2 - \frac{1}{4}\|y - s\|^2$ combined with the standard DCA framework [33,34].

4.2.1. Numerical experiments on a synthetic problem

Consider the following linear optimization problem

$$Q(c) = \min_x \left\{ c_1 x_1 + c_2 x_2 \,:\, 2x_1 + x_2 \geq 3, x_1 + 2x_2 \geq 3 \right\}.$$

Set the true cost vector to be $(c_1, c_2) = (2, 2)$. The corresponding optimal value is $\hat{v} = 4$, achieved at the optimal solution $\hat{x} = (1, 1)$. With the admissible set $C := \{c \in \mathbb{R}^2 : c_1 \geq 0,\ c_2 \geq 0\}$, the inverse optimal value problem is formulated as

$$\min_{c \in C} \left\{ d_i(Q(c), \hat{v}) \right\},$$

where $d_1(v, \hat{v}) := |v - \hat{v}|$ and $d_2(v, \hat{v}) := \frac{1}{2}(v - \hat{v})^2$. The initial penalty parameter is $\rho_0 = 10^3$, and the algorithm is started at the point $z^0 = (c^0, x^0, y^0, s^0)$ with

$$c^0 = (1, 1), \quad x^0 = (2, 2), \quad y^0 = \left(\tfrac{1}{3}, \tfrac{1}{3}\right), \quad s^0 = (3, 3).$$

This point satisfies $A^T y^0 = c^0$ and $Ax^0 - b = s^0$, so it is feasible for the polyhedron D in (5); however, y^0 and s^0 violate complementarity because $(y^0)^T s^0 = \frac{1}{3} \cdot 3 + \frac{1}{3} \cdot 3 = 2 > 0$. The algorithm therefore starts from a point at which the exact penalty is active and must steer the iterates toward complementarity while matching the observed optimal value $\hat{v} = 4$. Since $y^0 - s^0 = (-8/3, -8/3)$, we have $\Pi_{\mathbb{R}^2_+}(y^0 - s^0) = (0, 0)$ and $V^0 = (0, 0)$. The first subproblem, solved with the interior-point solver described in Section 4, reads

$$\begin{aligned} \min_z \quad & d_i(y^T b, \hat{v}) + \rho^0 e^T y - \rho^0 \left(e^T \Pi_{\mathbb{R}^m_+}(y^0 - s^0) + (V^0)^T (y - y^0) + (V^0)^T (s - s^0) \right) \\ s.t. \quad & A^T y = c, \\ & Ax - b = s, \\ & y \in \mathbb{R}^2_+,\ s \in \mathbb{R}^2_+,\ c_1 \geq 0,\ c_2 \geq 0. \end{aligned}$$

Because of the linearization of h_1, the first subproblem reduces to $\min d_i(y^T b, \hat{v}) + \rho_0 e^T y$ over the polyhedron; it is feasible (e.g., z^0 itself is feasible) and convex. When d_1 is used, the nonsmooth absolute value is modeled in epigraph form, so that every subproblem is a smooth convex program. The optimal solutions c^* and x^* are then obtained for the two distance functions d_1 and d_2. Table 2 reports the results together with those of the two comparison methods.

For the nonsmooth distance d_1, Algorithm 1 recovers the true cost vector to within 2.1×10^{-5} in 12 iterations, with value-matching residual $d = 2.9 \times 10^{-5}$ and $p(z^*) = 0$; the final point is $c^* \approx (2.0000, 2.0000)$, $x^* = (1, 1)$. For the smooth distance d_2 the final iterate again satisfies $p(z^*) = 0$ and is feasible for D, but the first-order balance between value matching and the proximal tie-breaking term leaves a value residual of about 10^{-2} ($c^* \approx (1.99, 1.99)$). This behavior is specific to smooth distance functions: for d_1 the balance is absorbed by the subdifferential of the absolute value, so the value residual stays at the level of the solver tolerance. The block-coordinate bilinear-penalty method BCD-BP reaches the same solutions in a few iterations on this two-dimensional example. The

Table 2
Numerical results on the synthetic problem: Algorithm 1 (Ours), the block-coordinate bilinear-penalty method (BCD-BP), and the generic DC algorithm on the bilinear-penalty model (DCA-BP).

Distance	Ours			BCD-BP			DCA-BP		
	Res	Iter	Time(s)	Res	Iter	Time(s)	Res	Iter	Time(s)
d_1	2.1×10^{-5}	12	0.84	2.3×10^{-5}	10	0.82	1.4×10^{0}	200[a]	30.7
d_2	1.4×10^{-2}	5	0.31	1.5×10^{-2}	4	0.30	8.0×10^{-2}	200[a]	19.2

[a] The step-size criterion was not met within 200 iterations.

Table 3
Numerical results on high-dimensional cases with the distance function d_1 under different sizes.

n	m	Ours			BCD-BP		
		Res	Iter	Time(s)	Res	Iter	Time(s)
5	100	1.0×10^{-4}	22	6.0	7.3×10^{-4}	19	4.0
	500	1.6×10^{-3}	26	11.1	1.8×10^{-3}	20	11.4
	1000	3.0×10^{-4}	23	34.7	2.7×10^{-4}	22	34.5
10	100	3.5×10^{-3}	28	8.8	1.5×10^{-3}	22	9.0
	500	2.5×10^{-3}	138	136.4	4.0×10^{-3}	173	162.7
	1000	4.8×10^{-3}	142	203.6	4.3×10^{-3}	185	212.5

Table 4
Numerical results on high-dimensional cases with the distance function d_2 under different sizes.

n	m	Ours			BCD-BP		
		Res	Iter	Time(s)	Res	Iter	Time(s)
5	100	9.1×10^{-4}	20	5.5	2.8×10^{-4}	17	5.6
	500	2.2×10^{-3}	24	10.9	1.7×10^{-3}	18	11.2
	1000	1.2×10^{-4}	22	30.8	3.0×10^{-4}	20	31.5
10	100	1.8×10^{-3}	25	8.6	2.2×10^{-3}	23	8.4
	500	2.6×10^{-3}	124	124.2	8.8×10^{-4}	152	140.7
	1000	4.2×10^{-3}	130	193.8	3.3×10^{-3}	169	201.4

generic DC method DCA-BP, by contrast, does not converge within 200 iterations and terminates at stationary points of the bilinear-penalty model with large value-matching residuals ($d = 2.0$ for d_1 and $d = 0.11$ for d_2), which illustrates the difficulty of the nonconvex bilinear penalty that motivates the exact ℓ_1 penalty of this paper.

4.2.2. Numerical results on high-dimensional cases

We generate random $m \times n$ matrices A whose entries are independently drawn from the standard normal distribution. To guarantee that the forward linear program is feasible and bounded below for every instance, we construct $b = Ax^* - s^0$ with a random x^* and a random vector $s^0 \geq 0$ (so that $x^* \in \{x : Ax \geq b\}$), and we choose the true cost vector $\bar{c} = A^T\bar{y}$ with a random $\bar{y} \geq 0$, which makes the dual feasible and hence the forward LP bounded below. The target value $\hat{v}$ is the optimal value $\min\{\bar{c}^T x : Ax \geq b\}$, computed with HiGHS. The admissible set is the box $C := \{c : \bar{c} - 0.1 \cdot l \leq c \leq \bar{c} + 0.1 \cdot u\}$, where l and u are random $n \times 1$ vectors with elements in $[0, 1]$. The initial point is $c^0 := \bar{c}$ with (x^0, y^0, s^0) an optimal primal–dual–slack triple of the forward problem, so that $z^0 \in D$ and $p(z^0) = 0$. Tables 3 and 4 report the results of Algorithm 1 and of BCD-BP for varying n and m.

On all 12 instances Algorithm 1 meets the step-size criterion within 20–142 iterations. The recovered cost vectors lie on the value-matching manifold, at distances between 1.0×10^{-4} and 4.8×10^{-3} from the true vector $\bar{c}$. The computation time grows with m (from 5.5 s to about 203.6 s), reflecting the cost of the m-dimensional quadratic subproblems solved by an interior-point method. The block-coordinate method BCD-BP requires 17–185 iterations on the same instances; it is faster than our method on some instances and slower on others, so neither method dominates across the test set.

4.3. Comparison with a recent DC-based method [30]

We further compare our method with the recent DC-based algorithm of Li and Cui [30] on a moderately challenging instance ($n = 10$, $m = 1000$). Since no default parameter values are given in [30], we set the augmented Lagrangian penalty parameter to 10^4 and the subproblem tolerance to 10^{-6}, and we terminated the method when the KKT residual fell below 10^{-2}. All results are averaged over 20 independent runs with identical initializations.

Table 5 reports the mean and standard deviation of residuals, CPU time, and iteration counts. The two methods are of comparable accuracy and cost on this instance: our method with d_2 attains a marginally lower residual (4.22×10^{-3} vs. 4.24×10^{-3}), with

Table 5
Comparison with a recent DC-based method [30].

Method	Res (10^{-3})	Time(s)	Iter
Ours (d_1)	4.85 ± 0.35	195.9 ± 3.8	144.2 ± 3.2
Ours (d_2)	4.22 ± 0.18	192.7 ± 6.2	131.6 ± 2.2
Li & Cui [30]	4.24 ± 0.32	194.5 ± 4.4	145.7 ± 2.3

Table 6
Statistics over 20 random initializations.

	Res-mean	Res-min	Res-max	d-max	p-max	Iter-mean	Time-mean
d_1	5.7×10^{-3}	1.3×10^{-3}	9.7×10^{-3}	2.2×10^{-7}	2.0×10^{-7}	26.2	11.5
d_2	6.3×10^{-3}	1.7×10^{-3}	8.9×10^{-3}	1.2×10^{-7}	2.1×10^{-7}	24.5	10.7

Table 7
Effect of ρ_0 on residual and iterations.

ρ_0	d_1			d_2		
	Res	d	Iter	Res	d	Iter
10	4.3×10^{-3}	6.7×10^{-9}	40	5.3×10^{-3}	4.2×10^{-9}	39
10^2	2.0×10^{-3}	5.5×10^{-8}	35	2.4×10^{-3}	5.4×10^{-8}	32
10^3	1.4×10^{-3}	2.1×10^{-7}	26	2.2×10^{-3}	1.2×10^{-7}	24
10^4	1.0×10^{-3}	8.5×10^{-7}	28	2.6×10^{-3}	3.3×10^{-7}	26
10^5	3.0×10^{-3}	1.8×10^{-5}	30	3.1×10^{-3}	1.7×10^{-5}	29
10^6	3.1×10^{-2}	1.4×10^{-4}	34	4.2×10^{-2}	1.4×10^{-4}	32
10^7	3.1×10^{-2}	1.2×10^{-3}	39	4.4×10^{-2}	1.3×10^{-3}	36
10^8	3.1×10^{-2}	1.1×10^{-3}	45	4.5×10^{-2}	1.2×10^{-3}	42

comparable CPU time (192.7 s vs. 194.5 s) and comparable iteration counts (131.6 vs. 145.7); these values are consistent with those reported for the same size in Tables 3 and 4. As an empirical diagnostic, we also monitored the conditioning of the quadratic subproblems solved at each iteration and observed that the Hessians associated with our subproblems were better conditioned than those arising in the general DC decomposition of Li and Cui [30]. We stress that this is an empirical observation on a single instance, and that we do not use it to claim a systematic advantage: on this instance the two methods perform comparably.

4.4. Global optimality assessment

To examine whether the algorithm converges to a global optimum, we run 20 random initializations, each constructed as a point of D: we sample c^0 uniformly from C (rejecting draws for which the forward LP with cost c^0 is unbounded below) and take (x^0, y^0, s^0) as an optimal primal–dual–slack triple of the forward problem, so that $A^T y^0 = c^0$ and $Ax^0 - b = s^0$ hold and the corresponding complementarity violation is zero. The instance is fixed with $n = 5$, $m = 500$. Table 6 summarizes the results.

All 20 runs converge to the value-matching manifold and meet the step-size criterion: the complementarity measure at the final iterate never exceeds 2.1×10^{-7}, and the value-matching residual is at most 2.2×10^{-7} for d_1 and 1.2×10^{-7} for d_2. Since the penalized objective $f = d_i + \rho p$ is nonnegative and vanishes at every solution of the IOVP, each final iterate satisfies $f \le 1.2 \times 10^{-7} + 10^3 \cdot 2.1 \times 10^{-7} \approx 2.1 \times 10^{-4}$, i.e., the stationary points obtained from all initializations are near-global minimizers of the penalized problem. The step-size criterion is met within 20–36 iterations for d_1 and 22–34 iterations for d_2 (means 26.2 and 24.5). The recovered cost vectors lie on the value-matching manifold at distances 1.3×10^{-3} to 9.7×10^{-3} from $\hat{c}$; this spread reflects the intrinsic freedom of the value-matching set of the IOVP, and the proximal term selects the point preferred by the initial estimate of each run.

4.5. Sensitivity to the penalty parameter

We fix the instance with $n = 5$, $m = 500$ and vary the initial penalty parameter ρ_0 from 10 to 10^8. Table 7 shows that the method is robust over a wide range of the initial penalty parameter: for $\rho_0 \in [10, 10^5]$ the recovered cost vector satisfies Res $\le 5.3 \times 10^{-3}$ and the value-matching residual d is at most 1.8×10^{-5}, whereas for $\rho_0 \ge 10^6$ the value-matching and proximal terms are scaled down by the factor $1/\rho$ in the subproblems and become numerically negligible relative to the linearized penalty. In the latter regime the recovered cost vector drifts along the value-matching manifold (Res grows to about 3.1×10^{-2} for d_1 and up to 4.5×10^{-2} for d_2) and the value-matching residual d grows as well (up to 1.3×10^{-3} in the tested range), while the iteration count varies only mildly with ρ_0 (24–45 over the whole range). Note that ρ_0 enters only the objective of the convex subproblems, so it does not affect their feasibility: every subproblem is feasible for every value of ρ_0. The default choice $\rho_0 = 10^3$ works well across all tested instances.

5. Concluding remarks

In this paper, we have developed an exact penalty difference-of-convex (DC) algorithm for solving inverse optimal value problems in linear programming. By incorporating the KKT conditions and strong duality of the forward LP, we reformulated the original bilevel problem as a single-level program with complementarity constraints. The complementarity condition is penalized in the objective through an exact ℓ_1-type penalty, whose DC decomposition yields convex subproblems over a fixed polyhedral feasible region. We proved that every limit point of the sequence generated by the proposed DCA is a stationary point of the penalized problem (and, when the objective is made strongly convex by a proximal term, every accumulation point is stationary), without relying on any constraint qualification, and we established the exactness of the penalty: under standard conditions (including a relative error bound or MPEC-LICQ), global minimizers of the penalized problem coincide with those of the IOVP, and the penalty is asymptotically exact in general. Unlike previous penalty-based approaches [27,28], our method provides both convex subproblems and a convergence guarantee to stationarity. The theoretical findings are supported by extensive numerical experiments on a synthetic example and high-dimensional instances, including comparisons with the bilinear-penalty reformulation approaches [27,28] and a recent DC-based method [30], a multiple-initialization study, and a sensitivity analysis with respect to the penalty parameter, which further confirm the robustness and scalability of our approach. Future work may focus on extending the penalty-type DC framework to other classes of inverse optimization problems, such as inverse conic or semidefinite programming.

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Data availability

No data was used for the research described in the article.

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